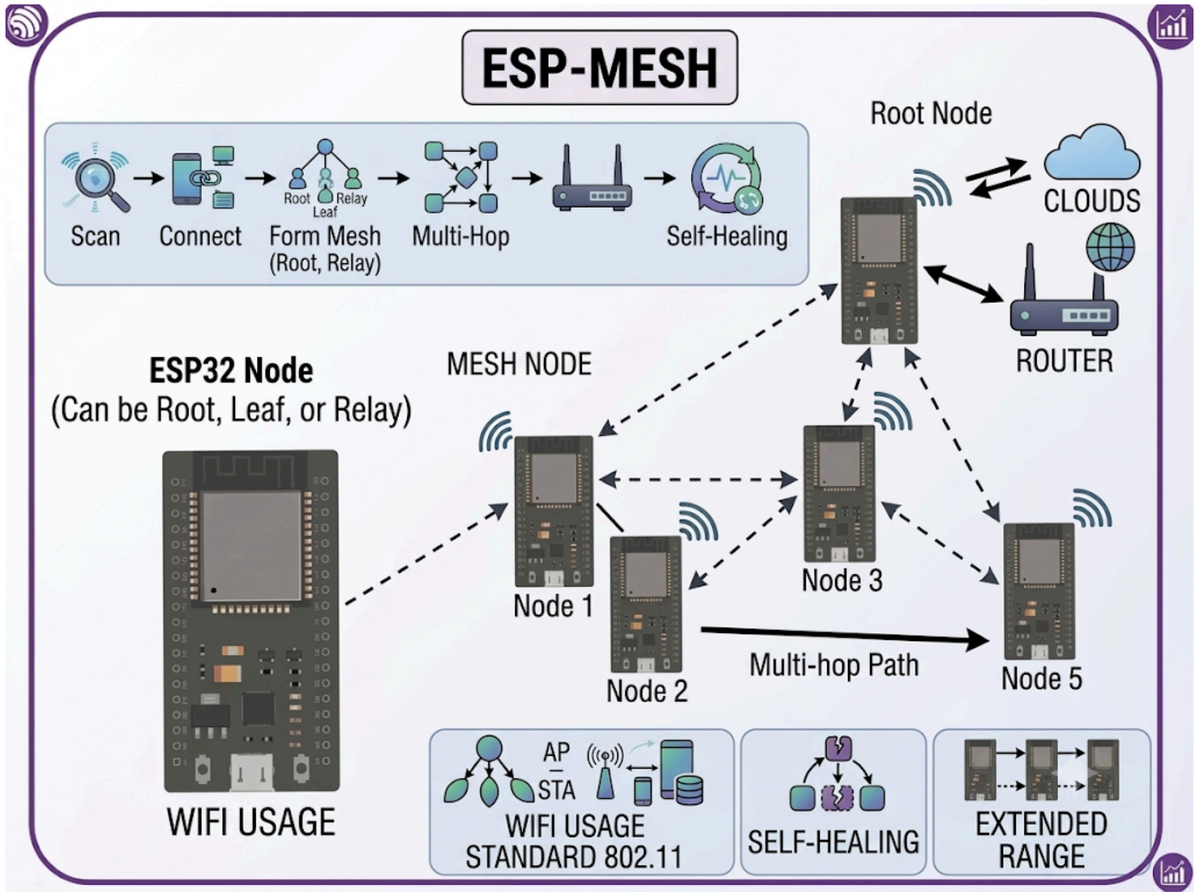
ESP MESH Protocol

Multi-hop, self-organizing mesh network built on top of Wi-Fi.

What Is ESP-MESH?

Multi-hop, self-organizing mesh network built on top of Wi-Fi.



Rides the Wi-Fi Radio

Uses standard Wi-Fi (802.11) softAP and Station modes to connect nodes dynamically.

No Access Point Needed

Formed automatically in a tree/mesh topology; a central router/AP is optional

Topology

Tree-based / Multi-hop mesh (Root Node, Leaf Nodes, and Node Relays).

Flexible Testing

Allows monitoring and testing of the whole mesh via a designated Root Node connected directly to a PC.

Where Is ESP MESH Used in IoT?

Replace peer-to-peer/sensor mesh cases

Large-scale IoT Networks

Extended-Range Networks

Self-Healing Infrastructure

Bi-directional Request-Response Services

Prerequisites & Hardware :

Hardware: 3 ESP32 development boards (3 nodes are recommended to demonstrate multi-hop routing and mesh self-healing).

USB data cable: Required for programming and Serial Monitor access. Make sure the cable supports data, not only charging.

Power: USB power is sufficient for normal testing. A separate regulated supply can be used if the board's power-input requirements are satisfied.

Software: Mention installing the painlessMesh library (or ESP-IDF esp_mesh) via the Arduino Library Manager alongside the core ESP32 board package

Network Requirements: A 2.4 GHz Wi-Fi Network (SSID and Password)

Step 1: Configure the Arduino IDE (Install painlessMesh and TaskScheduler libraries)

Step 2: Define Network Credentials (Set Mesh Prefix/SSID, Mesh Password, and Port).

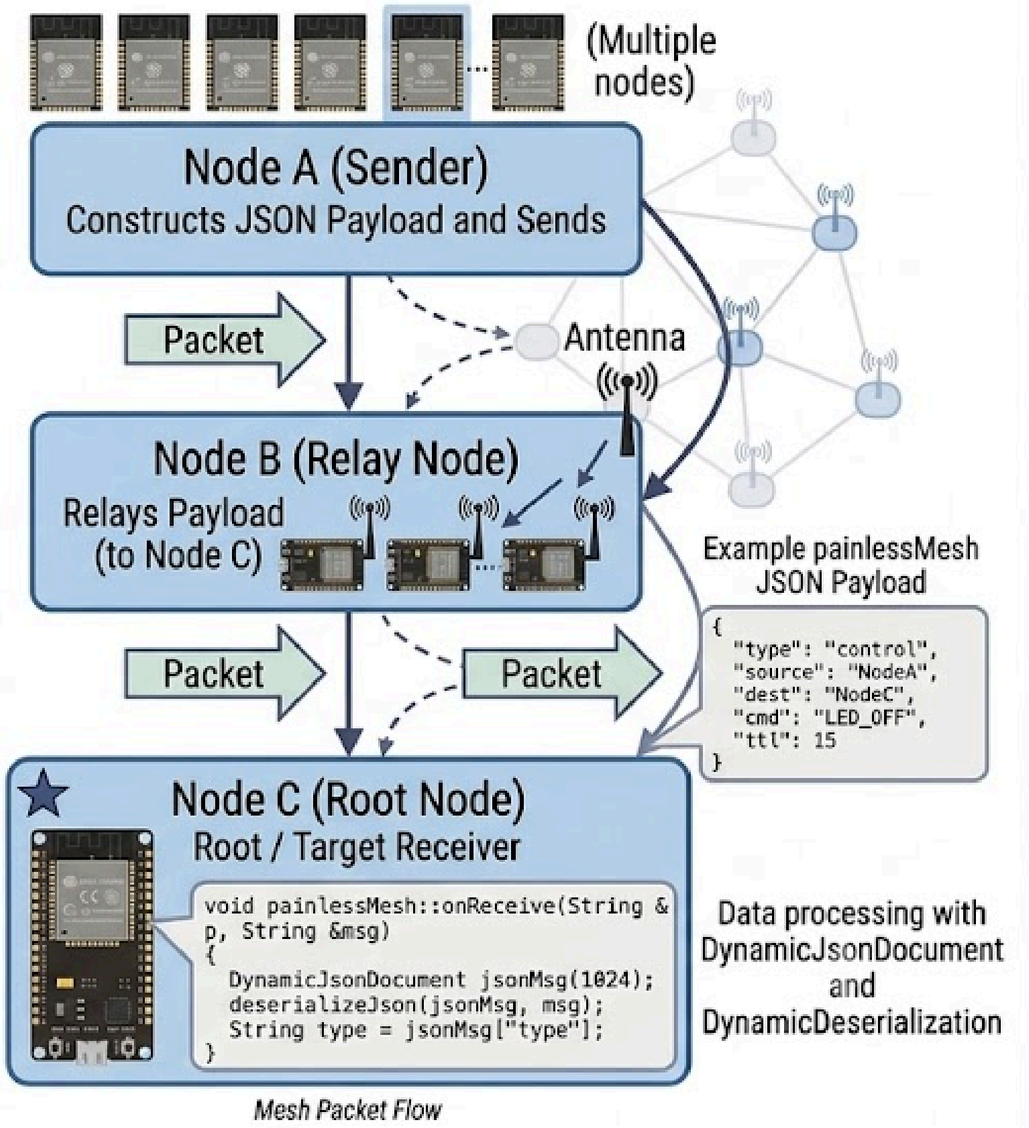
Step 3: Configure Node Initialization (mesh.init(), set node type, and attach event callbacks like onReceive).

Step 4: Set Up Task Scheduler

Step 5: Flash Nodes & Observe Self-Formation

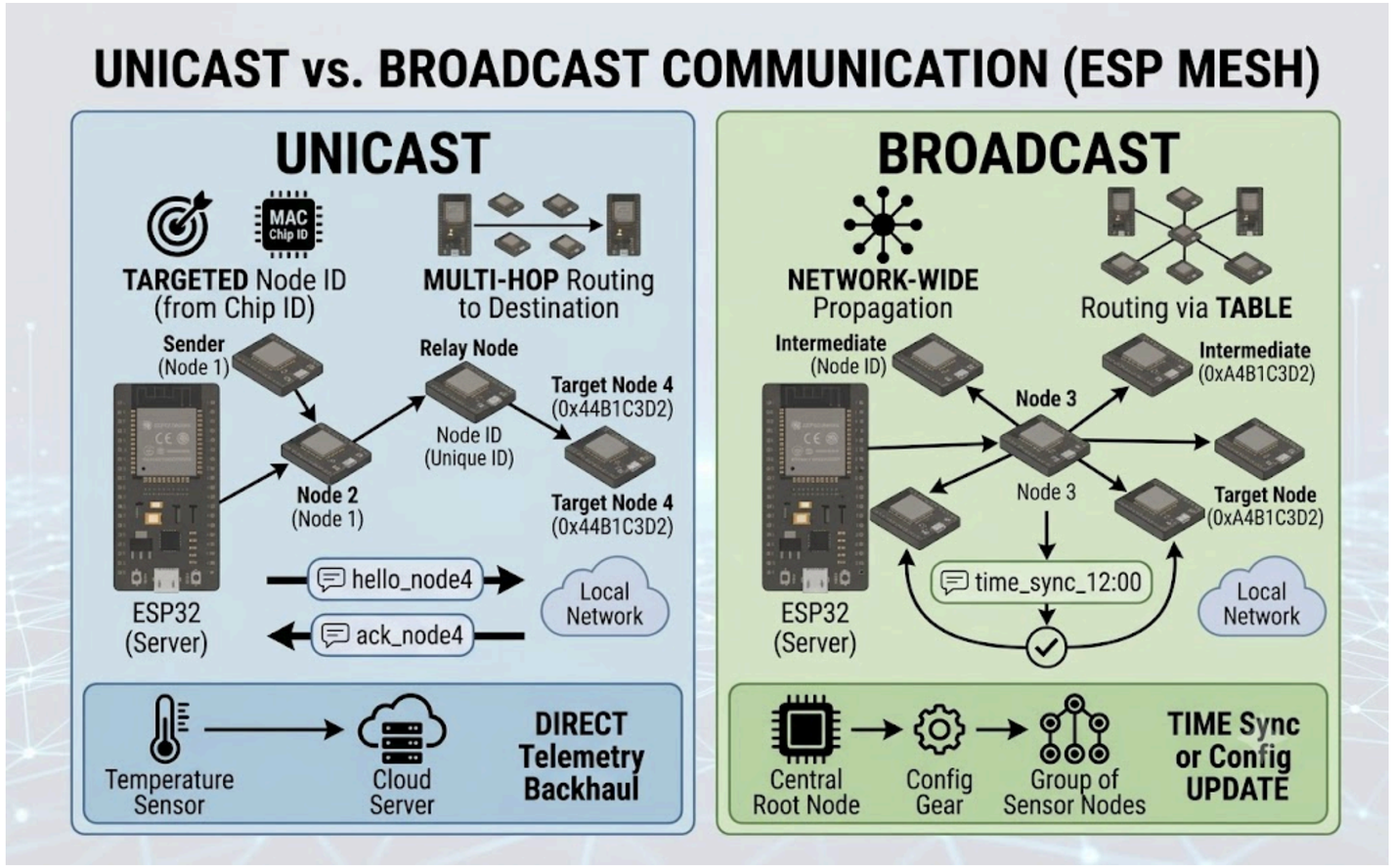
Step 6: Multi-hop Data Transmission

Understand the Multi-Hop Mesh Data Flow (painlessMesh)



ESP MESH Roles Comparison

In ESP-MESH, a Root Node (or Gateway) acts as the central hub listening for multi-hop traffic from internal mesh nodes, while a Node (Leaf/Relay) connects to the mesh network via Node IDs and mesh routing tables to send, receive, or relay structured data buffers.



An ESP-MESH connection relies on exact Node IDs (derived from the ESP32's MAC address) or broadcast routing rather than IP addresses, Ports, or URL paths. Ensure client node transmissions format payloads with expected message delimiters (such as JSON strings, \n, or structured bytes), as mesh reception callbacks depend on them to parse and route data buffers across intermediate nodes.

That's ESP MESH Communication!

The access the implementation guide
Scan the QR / click on the link in the description

Thank You